

Lecture 26

Swarm Intelligence

COMP532 – Machine Learning and Bioinspired Optimisation

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Lecture goals

- Introduce swarm intelligence as a **distributed search** paradigm.
- Explain how **local interaction** can generate useful **global behaviour**.
- Identify the core mechanisms: **self-organization**, **feedback**, **randomness**, and **stigmergy**.
- Prepare the ground for **Particle Swarm Optimisation (PSO)**, **Ant Colony Optimisation (ACO)**, and **swarm robotics**.

Why study swarm intelligence?

- Many real systems operate without a reliable central controller.
- Agents often have only **local observations** and **limited communication**.
- We want methods that are:
 - scalable,
 - robust to individual failure,
 - adaptive under change,
 - naturally parallel.
- Swarms show that useful collective behaviour can emerge from **simple local rules**.

What is swarm intelligence?

Working definition

Swarm intelligence studies how many relatively simple agents, using local information and limited interaction, can produce effective collective behaviour without central control.

- distributed system
- local interactions
- self-organization
- emergent global behaviour



<https://www.youtube.com/watch?v=2IAluwgAFD0>

A computational view

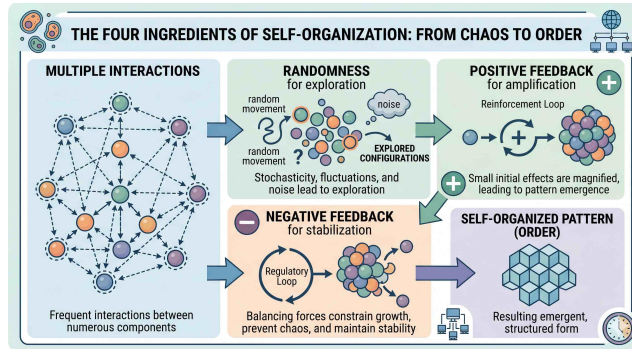
A swarm system typically has five ingredients

1. many agents
2. local state
3. local interaction
4. stochasticity / exploration
5. feedback and memory

The key question is not whether any one agent is clever, but whether the **population dynamics** are useful.

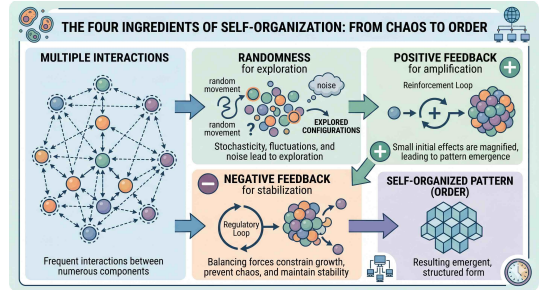
Self-organization

- Structure appears at the **global level** through interactions among lower-level components.
- Agents act using **purely local information**.
- The collective pattern is **not externally imposed**.
- Useful behaviour is therefore an **emergent property** of the system.



Four ingredients of self-organization

- Multiple interactions
- Randomness for exploration
- Positive feedback for amplification
- Negative feedback for stabilization



Interpretation

Good swarm systems must both **amplify useful discoveries** and **avoid runaway lock-in**.

Emergence: why simple rules can work

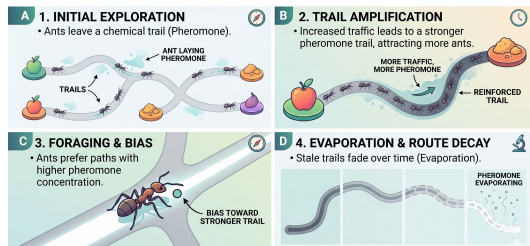
- Individuals do not need to understand the global objective.
- Repeated local reactions can create stable group-level structure.
- The intelligence may belong to the **collective process**, not to any single agent.

Key idea

In swarm systems, **coordination is often a property of interaction dynamics.**

Example 1: ants and stigmergy

- Ants can communicate **indirectly** through the environment.
- Pheromone trails bias future movement decisions.
- More traffic reinforces a trail.
- Evaporation weakens stale trails over time.



This is a powerful mechanism for distributed path selection.

What is stigmergy?

Definition

Stigmergy is indirect communication through modifications of the environment.

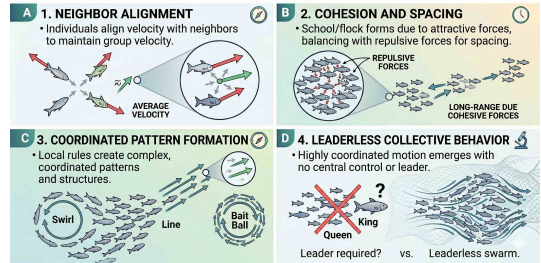
- Agents leave traces or markers.
- Those traces influence what later agents do.
- Memory is therefore **externalized into the environment**.

Preview

Ant Colony Optimisation will turn this idea into an algorithmic search mechanism.

Example 2: flocking and schooling

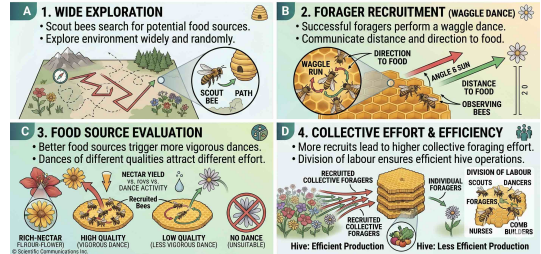
- Individuals align with neighbours.
- They maintain cohesion while avoiding collision.
- The result is highly coordinated motion with no leader required.



This motivates the idea that **social influence** can guide search.

Example 3: bees and recruitment

- Some bees explore widely for food.
- Successful foragers recruit others.
- Better food sources attract more collective effort.
- Division of labour improves efficiency.



This is a distributed mechanism for **resource allocation**.

Swarm intelligence as distributed problem solving

- no central planner
- parallel exploration
- adaptation to changing environments
- robustness to failure of individuals
- useful for optimization, search, routing, coordination, and robotics

Interpretation

Swarm intelligence is a natural model for **distributed problem solving under uncertainty**.

Strengths and limitations

Strengths

- scalable
- robust
- parallel
- adaptive

Limitations

- may converge slowly
- may lock into poor conventions
- behaviour can be hard to analyse
- sensitive to local rule design

A unifying summary

One idea to remember

Swarm intelligence is not about copying animals literally. It is about designing **distributed search and coordination mechanisms** from local interaction, memory, feedback, and randomness.

- PSO: social learning in continuous spaces
- ACO: stigmergic search in combinatorial spaces
- swarm robotics: embodied coordination under local sensing and communication